

SIH 2026 Master Project Handover Manual

Repository: github.com/soujhi/LegalMet_SIH | Scope: Complete system architecture, DoCA pipeline, credentials, and handover checklist

Problem Statement: SIH26036	Authority: DoCA / Legal Metrology	Test Suite: 100% Passing (test_flow.py)
Handover Date: 27 September 2026	Ingested Govt Models: 451 Certificates	Total Catalog: 454 Models in DB

1. Core Mission & Metrological Architecture (Layer A vs Layer B)

LegalMet Verify digitizes the statutory lifecycle for commercial weighing and measuring instruments in India. The system strictly decouples **Layer A: Government Model Approval Reference Data** (451 scraped DoCA certificates storing authoritative baseline tolerances, capacity, scale interval e, and approval marks) from **Layer B: Physical In-Situ Trader Deployments** (individual commercial scales registered with serial numbers, asset tags, GPS locations, and inspection histories). During field inspections, the deterministic rule engine verifies observations against the **Seventh Schedule, Part-II** and **OIML R 76-1** standards before issuing tamper-evident QR certificates.

2. Completed Deliverables & System Components

Subsystem	Component	Implementation Details & Deliverables
DoCA Ingestion	scripts/import_doca_model_approval.py	Extracted all 451 PDFs into detailed CSV, JSON, and SQLite catalog with 0 OCR fallbacks.
Backend APIs	backend/app/routers/ (11 modules)	FastAPI endpoints for Auth, Instruments, Applications, Scheduling, Verification, Rules, Certificates, Analytics, Audit.
Rule Engine	backend/app/services/rule_engine.py	Deterministic MPE evaluation ($\pm 0.5e$, $\pm 1.0e$, $\pm 1.5e$ initial; $\pm 1.0e$, $\pm 2.0e$, $\pm 3.0e$ in-service).
Certificates & QR	certificate_generator.py	ReportLab PDF generator with cryptographic SHA-256 fingerprint and tamper-evident QR code.
Frontend Portals	frontend/src/pages/ (15 views)	React 18 + TS + Tailwind portals for Trader, LMO Inspector, Admin Controller, and Public Verification.
Automated Tests	scripts/test_flow.py	100% passing Golden PASS (cert issuance) and Failure FAIL (MPE rejection) test suite.

3. Flagged Items for Refinement (Handover Checklist)

Item	Observed Behavior	Recommended Action	Target File
1. Class III Precedence	Substring check for 'CLASS III' matched 'CLASS IIII'.	Move Class IIII condition before Class III in if-else chain.	rule_engine.py
2. Scale Interval < 1g	e=0.1g on Class II balances skipped gram-to-kg conversion.	Explicitly divide by 1000 whenever unit is kg.	rule_engine.py
3. Cert Hash Keys	Dictionary key named 'model' vs 'model_series'.	Align hash dict keys between generator and router.	certificate_generator.py
4. UI Denominator Label	Catalog total shows 454 without subtitle.	Add label: '451 DoCA Certificates + 3 Seed Models'.	AdminDataQuality.tsx

4. Quick-Start Guide & Seed Credentials

Role	Name	Email Login	Password	Scope / Persona
Admin	Rajesh Verma	admin@legalmet.gov.in	Admin@123	State Inspector / Controller
LMO Officer	Amit Sharma	lmo.sharma@legalmet.gov.in	Lmo@123	Field Inspection Execution
Trader 1	Ramesh Patel	trader.patel@agrotraders.in	Trader@123	Patel Agro Commodities (Mandi)
Trader 2	Sanjay Gupta	trader.gupta@barhistore.in	Trader@123	Gupta Kirana Store (Retail)

5. Developer Commands to Run and Test

Backend Server: cd backend; pip install -r requirements.txt; python -m uvicorn app.main:app --reload --port 8000
Frontend App: cd frontend; npm install; npm run dev (Runs on http://localhost:5173 or :3000)
Automated Test Suite: python scripts/test_flow.py (Runs Golden PASS & FAIL flows end-to-end)

Repository: https://github.com/soujhi/LegalMet_SIH	Status: FULLY PUSHED & SYNCHRONIZED	Verdict: HANDOVER READY
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